

# SOLIDWORKS DimXpert API (.NET) — Reference

## (7/8): IDimXpertPlaneFeature- IDimXpertTangencyTolerance

SOLIDWORKS API Documentation (converted from CHM)

# IDimXpertPlaneFeature (Type)

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IDimXpertPlaneFeature Interface

[See Also](#) [Members](#) [Example](#)

[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertPlaneFeature Interface

Allows you to access specific information about DimXpert planes on the DimXpertManager tab of the Management Panel.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

NET Syntax

Visual Basic (Declaration)

Public Interface IDimXpertPlaneFeature

Visual Basic (Usage)

Dim instance As [IDimXpertPlaneFeature](#)

C#

public interface IDimXpertPlaneFeature

C++/CLI

public interface class IDimXpertPlaneFeature

Visual Basic for Applications (VBA) Syntax

See [DimXpertPlaneFeature](#).

Example

[Get DimXpert Feature Example \(VBA\)](#)  
[Get DimXpert Feature Example \(VB.NET\)](#)  
[Get DimXpert Features and Annotations Example \(C#\)](#)

Remarks

You can access more general DimXpert feature information using [IDimXpertFeature](#).

Accessors

[IDimXpertPart::GetFeatures](#) and [IDimXpertPart::!GetFeatures](#)  
[IDimXpertPart::GetFeature](#)  
[IDimXpertCompoundClosedSlot3dFeature::GetPlaneFeatures](#)  
[IDimXpertCompoundNotchFeature::GetPlaneFeatures](#)  
[IDimXpertCompoundWidthFeature::GetPlaneFeatures](#)  
[IDimXpertIntersectCircleFeature::GetPlaneFeature](#)  
[IDimXpertIntersectLineFeature::GetPlaneFeatures](#)  
[IFeature::GetSpecificFeature2](#)

See Also

[IDimXpertPlaneFeature Members](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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IDimXpertPlaneFeature Interface Methods

[See Also](#)

[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertPlaneFeature Interface

For a list of all members of this type, see [IDimXpertPlaneFeature members](#).

## Public Methods

| Name                                                                                                              | Description                                              |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
|  <a href="#">GetNominalPlane</a> | Gets the coordinates and vector for this DimXpert plane. |

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## See Also

[IDimXpertPlaneFeature Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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### IDimXpertPlaneFeature Interface Members

[See Also](#) [Methods](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertPlaneFeature Interface

The following tables list the members exposed by [IDimXpertPlaneFeature](#).

## Public Methods

| Name                                                                                                              | Description                                              |
|-------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
|  <a href="#">GetNominalPlane</a> | Gets the coordinates and vector for this DimXpert plane. |

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## See Also

[IDimXpertPlaneFeature Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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### GetNominalPlane Method (IDimXpertPlaneFeature)

[See Also](#) [Example](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPlaneFeature Interface](#) : GetNominalPlane Method (IDimXpertPlaneFeature)

Gets the coordinates and vector for this DimXpert plane.

## .NET Syntax

### Visual Basic (Declaration)

```
Function GetNominalPlane( _  
    ByRef X As System.Double, _  
    ByRef Y As System.Double, _  
    ByRef Z As System.Double, _  
    ByRef I As System.Double, _  
    ByRef J As System.Double, _  
    ByRef K As System.Double _  
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IDimXpertPlaneFeature  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim I As System.Double  
Dim J As System.Double  
Dim K As System.Double  
Dim value As System.Boolean  
  
value = instance.GetNominalPlane(X, Y, Z, I, J, K)
```

### C#

```
System.bool GetNominalPlane(  
    out System.double X,  
    out System.double Y,  
    out System.double Z,  
    out System.double I,
```

```
    out System.double J,  
    out System.double K  
)
```

C++/CLI

```
System.bool GetNominalPlane(  
& [Out] System.double X,  
& [Out] System.double Y,  
& [Out] System.double Z,  
& [Out] System.double I,  
& [Out] System.double J,  
& [Out] System.double K  
)
```

#### Parameters

X  
X-coordinate of the plane  
Y  
Y-coordinate of the plane  
Z  
Z-coordinate of the plane  
I  
I component of the direction vector of the plane  
J  
J component of the direction vector of the plane  
K  
K component of the direction vector of the plane

#### Return Value

True if method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPlaneFeature::GetNominalPlane](#).

#### Example

[Get DimXpert Feature Example \(VBA\)](#)  
[Get DimXpert Feature Example \(VB.NET\)](#)

#### See Also

[IDimXpertPlaneFeature Interface](#)  
[IDimXpertPlaneFeature Members](#)

#### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

# IDimXpertPositionTolerance (Type)

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IDimXpertPositionTolerance Interface

See Also [Members](#) [Example](#)

 

SolidWorks.Interop.swdimxpert Namespace : IDimXpertPositionTolerance Interface

Allows you to access DimXpert position tolerances.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

## .NET Syntax

### Visual Basic (Declaration)

Public Interface IDimXpertPositionTolerance

### Visual Basic (Usage)



Dim instance As [IDimXpertPositionTolerance](#)

### C#

public interface IDimXpertPositionTolerance

### C++/CLI

public interface class IDimXpertPositionTolerance

## Visual Basic for Applications (VBA) Syntax

See [DimXpertPositionTolerance](#).

## Example

[Get DimXpert Tolerance3 Example \(VBA\)](#)  
[Get DimXpert Tolerance3 Example \(VB.NET\)](#)  
[Get DimXpert Features and Annotations Example \(C#\)](#)

## Remarks

You can access more general geometric tolerance information using [IDimXpertTolerance](#).

## Accessors

[IDimXpertAnnotation::GetAppliedAnnotations](#) and [IDimXpertAnnotation::!GetAppliedAnnotations](#)  
[IDimXpertPart::GetAnnotations](#) and [IDimXpertPart::!GetAnnotations](#)  
[IDimXpertPart::GetAnnotation](#)  
[IDimXpertFeature::GetAppliedAnnotations](#) and [IDimXpertFeature::!GetAppliedAnnotations](#)  
[IFeature::GetSpecificFeature2](#)

## See Also

[IDimXpertPositionTolerance Members](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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IDimXpertPositionTolerance Interface Methods

See Also

 

SolidWorks.Interop.swdimxpert Namespace : IDimXpertPositionTolerance Interface

For a list of all members of this type, see [IDimXpertPositionTolerance members](#).

## Public Methods

| Name                                                                                                                | Description                                                      |
|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
|  <a href="#">GetMaxTolerance</a> | Gets the maximum tolerance for this DimXpert position tolerance. |

|                                                                                                                      |                                                                                       |
|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  <a href="#">GetPlanarZoneVector</a> | Gets the planar zone vector that is used to compute this DimXpert position tolerance. |
|  <a href="#">GetProjectedZone</a>   | Gets the projected zone value of this DimXpert position tolerance.                    |

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 See Also

[IDimXpertPositionTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**IDimXpertPositionTolerance Interface Members**  
[See Also](#) [Properties](#) [Methods](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : [IDimXpertPositionTolerance Interface](#)

The following tables list the members exposed by [IDimXpertPositionTolerance](#).

 Public Properties

| Name                                                                                                       | Description                                                                |
|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|  <a href="#">Modifier</a> | Gets the material condition modifier for this DimXpert position tolerance. |
|  <a href="#">ZoneType</a> | Gets the zone type of this DimXpert position tolerance.                    |

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 Public Methods

| Name                                                                                                                  | Description                                                                           |
|-----------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|  <a href="#">GetMaxTolerance</a>     | Gets the maximum tolerance for this DimXpert position tolerance.                      |
|  <a href="#">GetPlanarZoneVector</a> | Gets the planar zone vector that is used to compute this DimXpert position tolerance. |
|  <a href="#">GetProjectedZone</a>    | Gets the projected zone value of this DimXpert position tolerance.                    |

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 See Also

[IDimXpertPositionTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**IDimXpertPositionTolerance Interface Properties**  
[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : [IDimXpertPositionTolerance Interface](#)

For a list of all members of this type, see [IDimXpertPositionTolerance members](#).

 Public Properties

| Name                                                                                                         | Description                                                                |
|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|  <a href="#">Modifier</a> | Gets the material condition modifier for this DimXpert position tolerance. |
|  <a href="#">ZoneType</a> | Gets the zone type of this DimXpert position tolerance.                    |

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 See Also

[IDimXpertPositionTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**GetMaxTolerance Method (IDimXpertPositionTolerance)**  
[See Also](#) [Example](#)

  [SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPositionTolerance Interface](#) : [GetMaxTolerance Method \(IDimXpertPositionTolerance\)](#)

Gets the maximum tolerance for this DimXpert position tolerance.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetMaxTolerance( _  
    ByRef IsMax As System.Boolean, _
```

```
ByRef Tolerance As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPositionTolerance  
Dim IsMax As System.Boolean  
Dim Tolerance As System.Double  
Dim value As System.Boolean  
  
value = instance.GetMaxTolerance(IsMax, Tolerance)
```

C#

```
System.bool GetMaxTolerance(  
    out System.bool IsMax,  
    out System.double Tolerance  
)
```

C++/CLI

```
System.bool GetMaxTolerance(  
& [Out] System.bool IsMax,  
& [Out] System.double Tolerance  
)
```

#### Parameters

*IsMax*

True if maximum tolerance is set; false otherwise

*Tolerance*

Maximum tolerance value

#### Return Value

True if the method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPositionTolerance::GetMaxTolerance](#).

#### Example

[Get DimXpert Tolerance3 Example \(VBA\)](#)

[Get DimXpert Tolerance3 Example \(VB.NET\)](#)

#### See Also

[IDimXpertPositionTolerance Interface](#)

[IDimXpertPositionTolerance Members](#)

#### Availability

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**GetPlanarZoneVector Method (IDimXpertPositionTolerance)**

See Also [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPositionTolerance Interface](#) : GetPlanarZoneVector Method (IDimXpertPositionTolerance)

Gets the planar zone vector that is used to compute this DimXpert position tolerance.

#### .NET Syntax

Visual Basic (Declaration)

```
Function GetPlanarZoneVector( _
    ByRef I As System.Double, _
    ByRef J As System.Double, _
    ByRef K As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimXpertPositionTolerance
Dim I As System.Double
Dim J As System.Double
Dim K As System.Double
Dim value As System.Boolean

value = instance.GetPlanarZoneVector(I, J, K)
```

C#

```
System.bool GetPlanarZoneVector(
    out System.double I,
    out System.double J,
    out System.double K
)
```

C++/CLI

```
System.bool GetPlanarZoneVector(
    & [Out] System.double I,
    & [Out] System.double J,
    & [Out] System.double K
)
```

#### Parameters

*I*  
I component of the planar zone vector  
*J*  
J component of the planar zone vector  
*K*  
K component of the planar zone vector

#### Return Value

True if the method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPositionTolerance::GetPlanarZoneVector](#).

#### Example

[Get DimXpert Tolerance3 Example \(VBA\)](#)  
[Get DimXpert Tolerance3 Example \(VB.NET\)](#)

#### See Also

[IDimXpertPositionTolerance Interface](#)  
[IDimXpertPositionTolerance Members](#)

#### Availability

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**GetProjectedZone Method (IDimXpertPositionTolerance)**

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPositionTolerance Interface](#) : GetProjectedZone Method (IDimXpertPositionTolerance)

Gets the projected zone value of this DimXpert position tolerance.

## .NET Syntax

### Visual Basic (Declaration)

```
Function GetProjectedZone( _  
    ByRef Enabled As System.Boolean, _  
    ByRef Value As System.Double _  
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IDimXpertPositionTolerance  
Dim Enabled As System.Boolean  
Dim Value As System.Double  
Dim value As System.Boolean  
  
value = instance.GetProjectedZone(Enabled, Value)
```

### C#

```
System.bool GetProjectedZone(  
    out System.bool Enabled,  
    out System.double Value  
)
```

### C++/CLI

```
System.bool GetProjectedZone(  
& [Out] System.bool Enabled,  
& [Out] System.double Value  
)
```

#### Parameters

##### *Enabled*

True if enabled; false otherwise

##### *Value*

Projected zone value

#### Return Value

True if the method call is successful; false otherwise

## Visual Basic for Applications (VBA) Syntax

See [DimXpertPositionTolerance::GetProjectedZone](#).

## Example

[Get DimXpert Tolerance3 Example \(VBA\)](#)

[Get DimXpert Tolerance3 Example \(VB.NET\)](#)

## See Also

[IDimXpertPositionTolerance Interface](#)

[IDimXpertPositionTolerance Members](#)

## Availability

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**Modifier Property (IDimXpertPositionTolerance)**

See Also [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPositionTolerance Interface](#) : Modifier Property (IDimXpertPositionTolerance)

Gets the material condition modifier for this DimXpert position tolerance.

## .NET Syntax

#### Visual Basic (Declaration)

ReadOnly Property Modifier As [swDimXpertMaterialConditionModifier\\_e](#)

#### Visual Basic (Usage)



```
Dim instance As IDimXpertPositionTolerance  
Dim value As swDimXpertMaterialConditionModifier\_e  
  
value = instance.Modifier
```

#### C#

```
swDimXpertMaterialConditionModifier\_e Modifier {get;}
```

#### C++/CLI

```
property swDimXpertMaterialConditionModifier\_e Modifier {  
    swDimXpertMaterialConditionModifier\_e get();  
}
```

#### Property Value

Material condition modifier as defined in [swDimXpertMaterialConditionModifier\\_e](#)

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPositionTolerance::Modifier](#).

#### Example

[Get DimXpert Tolerance3 Example \(VBA\)](#)  
[Get DimXpert Tolerance3 Example \(VB.NET\)](#)

#### See Also

[IDimXpertPositionTolerance Interface](#)  
[IDimXpertPositionTolerance Members](#)

#### Availability

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**ZoneType Property (IDimXpertPositionTolerance)**

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertPositionTolerance Interface](#) : ZoneType Property (IDimXpertPositionTolerance)

Gets the zone type of this DimXpert position tolerance.

#### .NET Syntax

#### Visual Basic (Declaration)

ReadOnly Property ZoneType As [swDimXpertPositionZoneType\\_e](#)

#### Visual Basic (Usage)



```
Dim instance As IDimXpertPositionTolerance  
Dim value As swDimXpertPositionZoneType\_e  
  
value = instance.ZoneType
```

#### C#

```
swDimXpertPositionZoneType\_e ZoneType {get;}
```

#### C++/CLI

```
property swDimXpertPositionZoneType\_e ZoneType {  
    swDimXpertPositionZoneType\_e get();  
}
```

#### Property Value

Position zone type as defined in [swDimXpertPositionZoneType\\_e](#)

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertPositionTolerance::ZoneType](#).

#### Example

[Get DimXpert Tolerance3 Example \(VBA\)](#)  
[Get DimXpert Tolerance3 Example \(VB.NET\)](#)

#### See Also

[IDimXpertPositionTolerance Interface](#)  
[IDimXpertPositionTolerance Members](#)

#### Availability

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# IDimXpertRadiusDimTol (Type)

**IDimXpertRadiusDimTol** Interface  
[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertRadiusDimTol Interface

Allows you to access DimXpert radius dimension tolerances.  
**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

## .NET Syntax

Visual Basic (Declaration)

```
Public Interface IDimXpertRadiusDimTol
```

Visual Basic (Usage)



```
Dim instance As IDimXpertRadiusDimTol
```

C#

```
public interface IDimXpertRadiusDimTol
```

C++/CLI

```
public interface class IDimXpertRadiusDimTol
```

## Visual Basic for Applications (VBA) Syntax

See [DimXpertRadiusDimTol](#).

## Example

[Get DimXpert Tolerance2 Example \(VBA\)](#)  
[Get DimXpert Tolerance2 Example \(VB.NET\)](#)  
[Get DimXpert Features and Annotations in a Model Example \(C#\)](#)

## Remarks

Use the Dimension Size icon on the DimXpert toolbar to create radius dimensions on non-closed curved faces.  
You can access more general dimension tolerance information using [IDimXpertDimensionTolerance](#).

## Accessors

[IDimXpertAnnotation::GetAppliedAnnotations](#) and [IDimXpertAnnotation::!GetAppliedAnnotations](#)  
[IDimXpertPart::GetAnnotations](#) and [IDimXpertPart::!GetAnnotations](#)  
[IDimXpertPart::GetAnnotation](#)  
[IDimXpertFeature::GetAppliedAnnotations](#) and [IDimXpertFeature::!GetAppliedAnnotations](#)  
[IFeature::GetSpecificFeature2](#)

## See Also

[IDimXpertRadiusDimTol Members](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)  
[IDimXpertDiameterDimTol Interface](#)

**IDimXpertRadiusDimTol** Interface **Members**  
[See Also](#) [Properties](#)



[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertRadiusDimTol Interface

The following tables list the members exposed by [IDimXpertRadiusDimTol](#).

## Public Properties

| Name                                                                                                               | Description                                                              |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
|  <a href="#">PatternTreatment</a> | Gets the pattern treatment for this DimXpert radius dimension tolerance. |

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## See Also

[IDimXpertRadiusDimTol Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)  
[IDimXpertDiameterDimTol Interface](#)

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## IDimXpertRadiusDimTol Interface Properties

[See Also](#)

[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertRadiusDimTol Interface

For a list of all members of this type, see [IDimXpertRadiusDimTol members](#).

## Public Properties

| Name                                                                                                               | Description                                                              |
|--------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
|  <a href="#">PatternTreatment</a> | Gets the pattern treatment for this DimXpert radius dimension tolerance. |

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## See Also

[IDimXpertRadiusDimTol Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)  
[IDimXpertDiameterDimTol Interface](#)

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## PatternTreatment Property (IDimXpertRadiusDimTol)

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertRadiusDimTol Interface](#) : PatternTreatment Property (IDimXpertRadiusDimTol)

Gets the pattern treatment for this DimXpert radius dimension tolerance.

## NET Syntax

### Visual Basic (Declaration)

ReadOnly Property PatternTreatment As [swDimXpertPatternTreatmentType\\_e](#)

### Visual Basic (Usage)



```
Dim instance As IDimXpertRadiusDimTol
Dim value As swDimXpertPatternTreatmentType\_e

value = instance.PatternTreatment
```

### C#

```
swDimXpertPatternTreatmentType\_e PatternTreatment {get;}
```

### C++/CLI

```
property swDimXpertPatternTreatmentType\_e PatternTreatment {
    swDimXpertPatternTreatmentType\_e get();
}
```

## Property Value

Pattern treatment type as defined in [swDimXpertPatternTreatmentType\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [DimXpertRadiusDimTol::PatternTreatment](#).

## Example

[Get DimXpert Tolerance2 Example \(VBA\)](#)  
[Get DimXpert Tolerance2 Example \(VB.NET\)](#)

[Get DimXpert Features and Annotations in a Model Example \(C#\)](#)

 **See Also**

[IDimXpertRadiusDimTol Interface](#)  
[IDimXpertRadiusDimTol Members](#)

 **Availability**

SOLIDWORKS 2010 FCS, Revision Number 18.0

# IDimXpertRefPlaneFeature (Type)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertRefPlaneFeature Interface

Allows you to access specific information about DimXpert reference planes on the DimXpertManager tab of the Management Panel.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

 .NET Syntax

Visual Basic (Declaration)

Public Interface IDimXpertRefPlaneFeature

Visual Basic (Usage)  [Copy Code](#)

Dim instance As [IDimXpertRefPlaneFeature](#)

C#

public interface IDimXpertRefPlaneFeature

C++/CLI

public interface class IDimXpertRefPlaneFeature

 Visual Basic for Applications (VBA) Syntax

See [DimXpertRefPlaneFeature](#).

 Remarks

You can access more general DimXpert feature information using [IDimXpertFeature](#).

 Accessors

- [IDimXpertPart::GetFeatures](#) and [IDimXpertPart::IGetFeatures](#)
- [IDimXpertPart::GetFeature](#)
- [IDimXpertCompoundClosedSlot3dFeature::GetPlaneFeatures](#)
- [IDimXpertCompoundNotchFeature::GetPlaneFeatures](#)
- [IDimXpertCompoundWidthFeature::GetPlaneFeatures](#)
- [IDimXpertIntersectCircleFeature::GetPlaneFeature](#)
- [IDimXpertIntersectLineFeature::GetPlaneFeatures](#)
- [IFeature::GetSpecificFeature2](#)

 See Also

[IDimXpertRefPlaneFeature Members](#)

[SolidWorks.Interop.swdimxpert Namespace](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertRefPlaneFeature Interface

For a list of all members of this type, see [IDimXpertRefPlaneFeature members](#).

 Public Methods

| Name                                                                                                                | Description                                                        |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|  <a href="#">GetNominalPlane</a> | Gets the coordinates and vector for this DimXpert reference plane. |

## See Also

[IDimXpertRefPlaneFeature Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

SOLIDWORKS DimXpert API Help

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### IDimXpertRefPlaneFeature Interface Members

[See Also](#) [Methods](#)

[SolidWorks.Interop.swdimxpert Namespace](#) : [IDimXpertRefPlaneFeature Interface](#)

The following tables list the members exposed by [IDimXpertRefPlaneFeature](#).

## Public Methods

| Name                                                                                                              | Description                                                        |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|  <a href="#">GetNominalPlane</a> | Gets the coordinates and vector for this DimXpert reference plane. |

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## See Also

[IDimXpertRefPlaneFeature Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

SOLIDWORKS DimXpert API Help

[Send comments on this topic.](#)

### GetNominalPlane Method (IDimXpertRefPlaneFeature)

[See Also](#)

[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertRefPlaneFeature Interface](#) : [GetNominalPlane Method \(IDimXpertRefPlaneFeature\)](#)  
Gets the coordinates and vector for this DimXpert reference plane.

## NET Syntax

### Visual Basic (Declaration)

```
Function GetNominalPlane( _  
    ByRef X As System.Double, _  
    ByRef Y As System.Double, _  
    ByRef Z As System.Double, _  
    ByRef I As System.Double, _  
    ByRef J As System.Double, _  
    ByRef K As System.Double _  
) As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IDimXpertRefPlaneFeature  
Dim X As System.Double  
Dim Y As System.Double  
Dim Z As System.Double  
Dim I As System.Double  
Dim J As System.Double  
Dim K As System.Double  
Dim value As System.Boolean  
  
value = instance.GetNominalPlane(X, Y, Z, I, J, K)
```

### C#

```
System.bool GetNominalPlane(  
    out System.double X,  
    out System.double Y,  
    out System.double Z,  
    out System.double I,  
    out System.double J,  
    out System.double K  
)
```

### C++/CLI

```
System.bool GetNominalPlane(  
& [Out] System.double X,  
& [Out] System.double Y,  
& [Out] System.double Z,  
& [Out] System.double I,  
& [Out] System.double J,  
& [Out] System.double K  
)
```

#### Parameters

X  
X-coordinate of the plane  
Y  
Y-coordinate of the plane  
Z  
Z-coordinate of the plane  
I  
I component of the direction vector of the plane  
J  
J component of the direction vector of the plane  
K  
K component of the direction vector of the plane

#### Return Value

True if method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertRefPlaneFeature::GetNominalPlane](#).

#### See Also

[IDimXpertRefPlaneFeature Interface](#)  
[IDimXpertRefPlaneFeature Members](#)

#### Availability

SOLIDWORKS 2026 SP02, Revision Number 34.2

# IDimXpertSphereFeature (Type)

**IDimXpertSphereFeature Interface**  
[See Also](#) [Members](#) [Example](#)

 

SolidWorks.Interop.swdimxpert Namespace : IDimXpertSphereFeature Interface

Allows you to access specific information about DimXpert spheres on the DimXpertManager tab of the Management Panel.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

 **.NET Syntax**

Visual Basic (Declaration)

Public Interface IDimXpertSphereFeature

Visual Basic (Usage)

Dim instance As [IDimXpertSphereFeature](#)

C#

public interface IDimXpertSphereFeature

C++/CLI

public interface class IDimXpertSphereFeature

 **Visual Basic for Applications (VBA) Syntax**

See [DimXpertSphereFeature](#).

 **Example**

[Get DimXpert Sphere Feature Example \(VBA\)](#)  
[Get DimXpert Sphere Feature Example \(VB.NET\)](#)  
[Get DimXpert Features and Annotations Example \(C#\)](#)

 **Remarks**

You can access more general DimXpert feature information using [IDimXpertFeature](#).

 **Accessors**

[IDimXpertPart::GetFeatures](#) and [IDimXpertPart::!GetFeatures](#)  
[IDimXpertPart::GetFeature](#)  
[IFeature::GetSpecificFeature2](#)

 **See Also**

[IDimXpertSphereFeature Members](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

**IDimXpertSphereFeature Interface Methods**  
[See Also](#)

 

SolidWorks.Interop.swdimxpert Namespace : IDimXpertSphereFeature Interface

For a list of all members of this type, see [IDimXpertSphereFeature members](#).

 **Public Methods**

| Name                                                                                                                 | Description                                               |
|----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
|  <a href="#">GetNominalSphere</a> | Gets the coordinates and vector for this DimXpert sphere. |

 See Also

[IDimXpertSphereFeature Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

SOLIDWORKS DimXpert API Help [Send comments on this topic.](#)

**IDimXpertSphereFeature Interface Members**  
[See Also](#) [Properties](#) [Methods](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertSphereFeature Interface

The following tables list the members exposed by [IDimXpertSphereFeature](#).

 Public Properties

| Name                                                                                                    | Description                                 |
|---------------------------------------------------------------------------------------------------------|---------------------------------------------|
|  <a href="#">Inner</a> | Gets whether the sphere is a hole or a pin. |

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 Public Methods

| Name                                                                                                               | Description                                               |
|--------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|
|  <a href="#">GetNominalSphere</a> | Gets the coordinates and vector for this DimXpert sphere. |

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 See Also

[IDimXpertSphereFeature Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

SOLIDWORKS DimXpert API Help [Send comments on this topic.](#)

**IDimXpertSphereFeature Interface Properties**  
[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertSphereFeature Interface

For a list of all members of this type, see [IDimXpertSphereFeature members](#).

 Public Properties

| Name                                                                                                      | Description                                 |
|-----------------------------------------------------------------------------------------------------------|---------------------------------------------|
|  <a href="#">Inner</a> | Gets whether the sphere is a hole or a pin. |

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 See Also

[IDimXpertSphereFeature Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**GetNominalSphere Method (IDimXpertSphereFeature)**  
[See Also](#) [Example](#)

  [SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertSphereFeature Interface](#) : GetNominalSphere Method (IDimXpertSphereFeature)  
Gets the coordinates and vector for this DimXpert sphere.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetNominalSphere( _  
    ByRef R As System.Double, _  
    ByRef X As System.Double, _  
    ByRef Y As System.Double, _  
    ByRef Z As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)  [Copy Code](#)

```
Dim instance As IDimXpertSphereFeature  
Dim R As System.Double
```

```
Dim X As System.Double
Dim Y As System.Double
Dim Z As System.Double
Dim value As System.Boolean

value = instance.GetNominalSphere(R, X, Y, Z)
```

```
C#

System.bool GetNominalSphere(
    out System.double R,
    out System.double X,
    out System.double Y,
    out System.double Z
)
```

```
C++/CLI

System.bool GetNominalSphere(
& [Out] System.double R,
& [Out] System.double X,
& [Out] System.double Y,
& [Out] System.double Z
)
```

Parameters

R  
Radius of the sphere  
X  
X-coordinate of the origin of the sphere  
Y  
Y-coordinate of the origin of the sphere  
Z  
Z-coordinate of the origin of the sphere

Return Value

True if method call is successful; false otherwise

 Visual Basic for Applications (VBA) Syntax

See [DimXpertSphereFeature::GetNominalSphere](#).

 Example

[Get DimXpert Sphere Feature Example \(VBA\)](#)  
[Get DimXpert Sphere Feature Example \(VB.NET\)](#)

 See Also

[IDimXpertSphereFeature Interface](#)  
[IDimXpertSphereFeature Members](#)

 Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

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Inner Property (IDimXpertSphereFeature)

[See Also](#) [Example](#)



SolidWorks.Interop.swdimxpert Namespace > [IDimXpertSphereFeature Interface](#) : Inner Property (IDimXpertSphereFeature)

Gets whether the sphere is a hole or a pin.

 .NET Syntax

Visual Basic (Declaration)

ReadOnly Property Inner As System.Boolean

Visual Basic (Usage)

 Copy Code

```
Dim instance As IDimXpertSphereFeature  
Dim value As System.Boolean  
  
value = instance.Inner
```

C#

```
System.bool Inner {get;}
```

C++/CLI

```
property System.bool Inner {  
    System.bool get();  
}
```

#### Property Value

True if sphere is a hole; false if a pin

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertSphereFeature::Inner](#).

#### Example

[Get DimXpert Sphere Feature Example \(VBA\)](#)  
[Get DimXpert Sphere Feature Example \(VB.NET\)](#)

#### See Also

[IDimXpertSphereFeature Interface](#)  
[IDimXpertSphereFeature Members](#)

#### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

# IDimXpertStraightnessTolerance (Type)

 

SolidWorks.Interop.swdimxpert Namespace : IDimXpertStraightnessTolerance Interface

Allows you to access DimXpert straightness tolerances.  
**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

 .NET Syntax

Visual Basic (Declaration)

Public Interface IDimXpertStraightnessTolerance

Visual Basic (Usage)  Copy Code

Dim instance As [IDimXpertStraightnessTolerance](#)

C#

public interface IDimXpertStraightnessTolerance

C++/CLI

public interface class IDimXpertStraightnessTolerance

 Visual Basic for Applications (VBA) Syntax

See [DimXpertStraightnessTolerance](#).

 Example

[Get DimXpert Tolerance4 Example \(VBA\)](#)  
[Get DimXpert Tolerance4 Example \(VB.NET\)](#)  
[Get DimXpert Features and Annotations Example \(C#\)](#)

 Remarks

You can access more general geometric tolerance information using [IDimXpertTolerance](#).

 Accessors

[IDimXpertAnnotation::GetAppliedAnnotations](#) and [IDimXpertAnnotation::!GetAppliedAnnotations](#)  
[IDimXpertPart::GetAnnotations](#) and [IDimXpertPart::!GetAnnotations](#)  
[IDimXpertPart::GetAnnotation](#)  
[IDimXpertFeature::GetAppliedAnnotations](#) and [IDimXpertFeature::!GetAppliedAnnotations](#)  
[IFeature::GetSpecificFeature2](#)

 See Also

[IDimXpertStraightnessTolerance Members](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

 

SolidWorks.Interop.swdimxpert Namespace : IDimXpertStraightnessTolerance Interface

For a list of all members of this type, see [IDimXpertStraightnessTolerance members](#).

 Public Methods

| Name                                                                                                                | Description                                                          |
|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
|  <a href="#">GetMaxTolerance</a> | Gets the maximum tolerance for this DimXpert straightness tolerance. |

|                                                                                                                       |                                                                                           |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|  <a href="#">GetPerUnitLength</a>     | Gets the per unit length for this DimXpert straightness tolerance.                        |
|  <a href="#">GetPlanarZoneVector</a> | Gets the planar zone vector that is used to compute this DimXpert straightness tolerance. |

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 See Also

[IDimXpertStraightnessTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

SOLIDWORKS DimXpert API Help [Send comments on this topic.](#)

**IDimXpertStraightnessTolerance Interface Members**

[See Also](#) [Properties](#) [Methods](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertStraightnessTolerance Interface

The following tables list the members exposed by [IDimXpertStraightnessTolerance](#).

 Public Properties

| Name                                                                                                       | Description                                                                    |
|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
|  <a href="#">Modifier</a> | Gets the material condition modifier for this DimXpert straightness tolerance. |
|  <a href="#">ZoneType</a> | Gets the zone type of this DimXpert straightness tolerance.                    |

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 Public Methods

| Name                                                                                                                  | Description                                                                               |
|-----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
|  <a href="#">GetMaxTolerance</a>     | Gets the maximum tolerance for this DimXpert straightness tolerance.                      |
|  <a href="#">GetPerUnitLength</a>    | Gets the per unit length for this DimXpert straightness tolerance.                        |
|  <a href="#">GetPlanarZoneVector</a> | Gets the planar zone vector that is used to compute this DimXpert straightness tolerance. |

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 See Also

[IDimXpertStraightnessTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**IDimXpertStraightnessTolerance Interface Properties**

[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertStraightnessTolerance Interface

For a list of all members of this type, see [IDimXpertStraightnessTolerance members](#).

 Public Properties

| Name                                                                                                         | Description                                                                    |
|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
|  <a href="#">Modifier</a> | Gets the material condition modifier for this DimXpert straightness tolerance. |
|  <a href="#">ZoneType</a> | Gets the zone type of this DimXpert straightness tolerance.                    |

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 See Also

[IDimXpertStraightnessTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**GetMaxTolerance Method (IDimXpertStraightnessTolerance)**

[See Also](#)

  [SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertStraightnessTolerance Interface](#) : GetMaxTolerance Method (IDimXpertStraightnessTolerance)

Gets the maximum tolerance for this DimXpert straightness tolerance.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetMaxTolerance( _  
    ByRef IsMax As System.Boolean, _
```

```
ByRef Tolerance As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimXpertStraightnessTolerance  
Dim IsMax As System.Boolean  
Dim Tolerance As System.Double  
Dim value As System.Boolean  
  
value = instance.GetMaxTolerance(IsMax, Tolerance)
```

C#

```
System.bool GetMaxTolerance(  
    out System.bool IsMax,  
    out System.double Tolerance  
)
```

C++/CLI

```
System.bool GetMaxTolerance(  
& [Out] System.bool IsMax,  
& [Out] System.double Tolerance  
)
```

#### Parameters

*IsMax*

True if maximum tolerance is set; false otherwise

*Tolerance*

Maximum tolerance value

#### Return Value

True if the method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertStraightnessTolerance::GetMaxTolerance](#).

#### See Also

[IDimXpertStraightnessTolerance Interface](#)

[IDimXpertStraightnessTolerance Members](#)

#### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

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**GetPerUnitLength Method (IDimXpertStraightnessTolerance)**

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertStraightnessTolerance Interface](#) : GetPerUnitLength Method (IDimXpertStraightnessTolerance)

Gets the per unit length for this DimXpert straightness tolerance.

#### .NET Syntax

Visual Basic (Declaration)

```
Function GetPerUnitLength( _  
    ByRef Enabled As System.Boolean, _  
    ByRef Distance As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimXpertStraightnessTolerance  
Dim Enabled As System.Boolean  
Dim Distance As System.Double
```

```
Dim value As System.Boolean

value = instance.GetPerUnitLength(Enabled, Distance)
```

C#

```
System.bool GetPerUnitLength(
    out System.bool Enabled,
    out System.double Distance
)
```

C++/CLI

```
System.bool GetPerUnitLength(
    & [Out] System.bool Enabled,
    & [Out] System.double Distance
)
```

#### Parameters

*Enabled*

True if per unit length is in effect; false otherwise

*Distance*

Length distance per unit length

#### Return Value

True if the method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertStraightnessTolerance::GetPerUnitLength](#).

#### Example

[Get DimXpert Tolerance4 Example \(VBA\)](#)

[Get DimXpert Tolerance4 Example \(VB.NET\)](#)

#### See Also

[IDimXpertStraightnessTolerance Interface](#)

[IDimXpertStraightnessTolerance Members](#)

#### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS DimXpert API Help

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**GetPlanarZoneVector Method (IDimXpertStraightnessTolerance)**

See Also [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertStraightnessTolerance Interface](#) : GetPlanarZoneVector Method (IDimXpertStraightnessTolerance)

Gets the planar zone vector that is used to compute this DimXpert straightness tolerance.

#### .NET Syntax

Visual Basic (Declaration)

```
Function GetPlanarZoneVector( _
    ByRef I As System.Double, _
    ByRef J As System.Double, _
    ByRef K As System.Double _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IDimXpertStraightnessTolerance
Dim I As System.Double
Dim J As System.Double
Dim K As System.Double
```

```
Dim value As System.Boolean  
value = instance.GetPlanarZoneVector(I, J, K)
```

C#

```
System.bool GetPlanarZoneVector(  
    out System.double I,  
    out System.double J,  
    out System.double K  
)
```

C++/CLI

```
System.bool GetPlanarZoneVector(  
    & [Out] System.double I,  
    & [Out] System.double J,  
    & [Out] System.double K  
)
```

#### Parameters

*I*  
I component of the planar zone vector  
*J*  
J component of the planar zone vector  
*K*  
K component of the planar zone vector

#### Return Value

True if the method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertStraightnessTolerance::GetPlanarZoneVector](#).

#### Example

[Get DimXpert Tolerance4 Example \(VBA\)](#)  
[Get DimXpert Tolerance4 Example \(VB.NET\)](#)

#### See Also

[IDimXpertStraightnessTolerance Interface](#)  
[IDimXpertStraightnessTolerance Members](#)

#### Availability

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**Modifier Property (IDimXpertStraightnessTolerance)**

See Also [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertStraightnessTolerance Interface](#) : Modifier Property (IDimXpertStraightnessTolerance)

Gets the material condition modifier for this DimXpert straightness tolerance.

#### .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Modifier As swDimXpertMaterialConditionModifier\_e
```

Visual Basic (Usage)



```
Dim instance As IDimXpertStraightnessTolerance  
Dim value As swDimXpertMaterialConditionModifier\_e  
  
value = instance.Modifier
```

C#

```
swDimXpertMaterialConditionModifier_e Modifier {get;}
```

C++/CLI

```
property swDimXpertMaterialConditionModifier_e Modifier {  
    swDimXpertMaterialConditionModifier_e get();  
}
```

#### Property Value

Material condition modifier as defined in [swDimXpertMaterialConditionModifier\\_e](#)

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertStraightnessTolerance::Modifier](#).

#### Example

[Get DimXpert Tolerance4 Example \(VBA\)](#)

[Get DimXpert Tolerance4 Example \(VB.NET\)](#)

#### See Also

[IDimXpertStraightnessTolerance Interface](#)

[IDimXpertStraightnessTolerance Members](#)

#### Availability

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**ZoneType Property (IDimXpertStraightnessTolerance)**

[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertStraightnessTolerance Interface](#) : ZoneType Property (IDimXpertStraightnessTolerance)

Gets the zone type of this DimXpert straightness tolerance.

#### .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property ZoneType As swDimXpertStraightnessZoneType_e
```

Visual Basic (Usage)



```
Dim instance As IDimXpertStraightnessTolerance
```

```
Dim value As swDimXpertStraightnessZoneType_e
```

```
value = instance.ZoneType
```

C#

```
swDimXpertStraightnessZoneType_e ZoneType {get;}
```

C++/CLI

```
property swDimXpertStraightnessZoneType_e ZoneType {  
    swDimXpertStraightnessZoneType_e get();  
}
```

#### Property Value

Position zone type as defined in [swDimXpertStraightnessZoneType\\_e](#)

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertStraightnessTolerance::ZoneType](#).

#### See Also

[IDimXpertStraightnessTolerance Interface](#)  
[IDimXpertStraightnessTolerance Members](#)

#### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

# IDimXpertSurfaceProfileTolerance (Type)

[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertSurfaceProfileTolerance Interface

Allows you to access DimXpert surface profile tolerances.  
**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

Public Interface IDimXpertSurfaceProfileTolerance

Visual Basic (Usage) Copy Code

Dim instance As [IDimXpertSurfaceProfileTolerance](#)

C#

public interface IDimXpertSurfaceProfileTolerance

C++/CLI

public interface class IDimXpertSurfaceProfileTolerance

Visual Basic for Applications (VBA) Syntax

See [DimXpertSurfaceProfileTolerance](#).

Example

[Get DimXpert Tolerance1 Example \(VBA\)](#)  
[Get DimXpert Tolerance1 Example \(VB.NET\)](#)  
[Get DimXpert Features and Annotations Example \(C#\)](#)

Remarks

You can access more general geometric tolerance information using [IDimXpertTolerance](#).

Accessors

[IDimXpertAnnotation::GetAppliedAnnotations](#) and [IDimXpertAnnotation::!GetAppliedAnnotations](#)  
[IDimXpertPart::GetAnnotations](#) and [IDimXpertPart::!GetAnnotations](#)  
[IDimXpertPart::GetAnnotation](#)  
[IDimXpertFeature::GetAppliedAnnotations](#) and [IDimXpertFeature::!GetAppliedAnnotations](#)  
[IFeature::GetSpecificFeature2](#)

See Also

[IDimXpertSurfaceProfileTolerance Members](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertSurfaceProfileTolerance Interface

The following tables list the members exposed by [IDimXpertSurfaceProfileTolerance](#).

Public Properties

| Name | Description |
|------|-------------|
|------|-------------|

|                                                                                                                      |                                                                    |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|  <a href="#">OuterToleranceValue</a> | Gets the outer tolerance value for this surface profile tolerance. |
|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|

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 See Also

[IDimXpertSurfaceProfileTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

SOLIDWORKS DimXpert API Help [Send comments on this topic.](#)

**IDimXpertSurfaceProfileTolerance Interface Properties**  
[See Also](#)



[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertSurfaceProfileTolerance Interface

For a list of all members of this type, see [IDimXpertSurfaceProfileTolerance members](#).

 Public Properties

| Name                                                                                                                  | Description                                                        |
|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
|  <a href="#">OuterToleranceValue</a> | Gets the outer tolerance value for this surface profile tolerance. |

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 See Also

[IDimXpertSurfaceProfileTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

SOLIDWORKS DimXpert API Help [Send comments on this topic.](#)

**OuterToleranceValue Property (IDimXpertSurfaceProfileTolerance)**  
[See Also](#) [Example](#)

 

[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertSurfaceProfileTolerance Interface](#) : OuterToleranceValue Property (IDimXpertSurfaceProfileTolerance)

Gets the outer tolerance value for this surface profile tolerance.

 .NET Syntax

|                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------------------------------------------|
| Visual Basic (Declaration)                                                                                                                   |
| ReadOnly Property OuterToleranceValue As System.Double                                                                                       |
| Visual Basic (Usage)  <a href="#">Copy Code</a>         |
| <pre>Dim instance As <a href="#">IDimXpertSurfaceProfileTolerance</a> Dim value As System.Double  value = instance.OuterToleranceValue</pre> |
| C#                                                                                                                                           |
| <pre>System.double OuterToleranceValue {get;}</pre>                                                                                          |
| C++/CLI                                                                                                                                      |
| <pre>property System.double OuterToleranceValue {     System.double get(); }</pre>                                                           |

**Property Value**

Outer tolerance value

 Visual Basic for Applications (VBA) Syntax

See [DimXpertSurfaceProfileTolerance.OuterToleranceValue](#).

 Example

[Get DimXpert Tolerance Example \(VBA\)](#)  
[Get DimXpert Tolerance Example \(VB.NET\)](#)

#### See Also

[IDimXpertSurfaceProfileTolerance Interface](#)  
[IDimXpertSurfaceProfileTolerance Members](#)

#### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

# IDimXpertSymmetryTolerance (Type)

 ▾

SolidWorks.Interop.swdimxpert Namespace : IDimXpertSymmetryTolerance Interface

Allows you to access DimXpert symmetry tolerances.  
**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

 .NET Syntax

Visual Basic (Declaration)

Public Interface IDimXpertSymmetryTolerance

Visual Basic (Usage)  Copy Code

Dim instance As [IDimXpertSymmetryTolerance](#)

C#

public interface IDimXpertSymmetryTolerance

C++/CLI

public interface class IDimXpertSymmetryTolerance

 Visual Basic for Applications (VBA) Syntax

See [DimXpertSymmetryTolerance](#).

 Example

- [Get DimXpert Tolerance2 Example \(VBA\)](#)
- [Get DimXpert Tolerance2 Example \(VB.NET\)](#)
- [Get DimXpert Features and Annotations Example \(C#\)](#)

 Remarks

You can access more general geometric tolerance information using [IDimXpertTolerance](#).

 Accessors

- [IDimXpertAnnotation::GetAppliedAnnotations](#) and [IDimXpertAnnotation::!GetAppliedAnnotations](#)
- [IDimXpertPart::GetAnnotations](#) and [IDimXpertPart::!GetAnnotations](#)
- [IDimXpertPart::GetAnnotation](#)
- [IDimXpertFeature::GetAppliedAnnotations](#) and [IDimXpertFeature::!GetAppliedAnnotations](#)
- [IFeature::GetSpecificFeature2](#)

 See Also

- [IDimXpertSymmetryTolerance Members](#)
- [SolidWorks.Interop.swdimxpert Namespace](#)

 ▾

SolidWorks.Interop.swdimxpert Namespace : IDimXpertSymmetryTolerance Interface

For a list of all members of this type, see [IDimXpertSymmetryTolerance members](#).

 Public Methods

| Name                                                                                                                | Description                                                      |
|---------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
|  <a href="#">GetMaxTolerance</a> | Gets the maximum tolerance for this DimXpert symmetry tolerance. |

|                                                                                                                   |                                                              |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|
|  <a href="#">GetZoneDirection</a> | Gets the zone direction of this DimXpert symmetry tolerance. |
|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------|

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 See Also

[IDimXpertSymmetryTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

SOLIDWORKS DimXpert API Help [Send comments on this topic.](#)

**IDimXpertSymmetryTolerance Interface Members**  
[See Also](#) [Properties](#) [Methods](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertSymmetryTolerance Interface

The following tables list the members exposed by [IDimXpertSymmetryTolerance](#).

 Public Properties

| Name                                                                                                       | Description                                                                |
|------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|  <a href="#">Modifier</a> | Gets the material condition modifier for this Dimxpert symmetry tolerance. |

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 Public Methods

| Name                                                                                                               | Description                                                      |
|--------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|
|  <a href="#">GetMaxTolerance</a>  | Gets the maximum tolerance for this DimXpert symmetry tolerance. |
|  <a href="#">GetZoneDirection</a> | Gets the zone direction of this DimXpert symmetry tolerance.     |

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 See Also

[IDimXpertSymmetryTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

SOLIDWORKS DimXpert API Help [Send comments on this topic.](#)

**IDimXpertSymmetryTolerance Interface Properties**  
[See Also](#)

 [SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertSymmetryTolerance Interface

For a list of all members of this type, see [IDimXpertSymmetryTolerance members](#).

 Public Properties

| Name                                                                                                         | Description                                                                |
|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
|  <a href="#">Modifier</a> | Gets the material condition modifier for this Dimxpert symmetry tolerance. |

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 See Also

[IDimXpertSymmetryTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**GetMaxTolerance Method (IDimXpertSymmetryTolerance)**  
[See Also](#)

  [SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertSymmetryTolerance Interface](#) : GetMaxTolerance Method (IDimXpertSymmetryTolerance)

Gets the maximum tolerance for this DimXpert symmetry tolerance.

 .NET Syntax

|                                                                                                                                                                                    |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Visual Basic (Declaration)                                                                                                                                                         |
| <pre>Function GetMaxTolerance( _<br/>    ByRef <a href="#">IsMax</a> As System.Boolean, _<br/>    ByRef <a href="#">Tolerance</a> As System.Double _<br/>) As System.Boolean</pre> |
| Visual Basic (Usage)  Copy Code                                                               |

```

Dim instance As IDimXpertSymmetryTolerance
Dim IsMax As System.Boolean
Dim Tolerance As System.Double
Dim value As System.Boolean

value = instance.GetMaxTolerance(IsMax, Tolerance)

```

C#

```

System.bool GetMaxTolerance(
    out System.bool IsMax,
    out System.double Tolerance
)

```

C++/CLI

```

System.bool GetMaxTolerance(
& [Out] System.bool IsMax,
& [Out] System.double Tolerance
)

```

#### Parameters

*IsMax*

True if maximum tolerance is set; false otherwise

*Tolerance*

Maximum tolerance value

#### Return Value

True if the method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertSymmetryTolerance::GetMaxTolerance](#).

#### See Also

[IDimXpertSymmetryTolerance Interface](#)  
[IDimXpertSymmetryTolerance Members](#)

#### Availability

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**GetZoneDirection Method (IDimXpertSymmetryTolerance)**

See Also [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertSymmetryTolerance Interface](#) : GetZoneDirection Method (IDimXpertSymmetryTolerance)

Gets the zone direction of this DimXpert symmetry tolerance.

#### .NET Syntax

Visual Basic (Declaration)

```

Function GetZoneDirection( _
    ByRef I As System.Double, _
    ByRef J As System.Double, _
    ByRef K As System.Double _
) As System.Boolean

```

Visual Basic (Usage)



```

Dim instance As IDimXpertSymmetryTolerance
Dim I As System.Double
Dim J As System.Double
Dim K As System.Double
Dim value As System.Boolean

```

```
value = instance.GetZoneDirection(I, J, K)
```

C#

```
System.bool GetZoneDirection(  
    out System.double I,  
    out System.double J,  
    out System.double K  
)
```

C++/CLI

```
System.bool GetZoneDirection(  
& [Out] System.double I,  
& [Out] System.double J,  
& [Out] System.double K  
)
```

#### Parameters

*I*  
I component of the zone direction vector  
*J*  
J component of the zone direction vector  
*K*  
K component of the zone direction vector

#### Return Value

True if the method call is successful; false otherwise

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertSymmetryTolerance::GetZoneDirection](#).

#### Example

[Get DimXpert Tolerance2 Example \(VBA\)](#)  
[Get DimXpert Tolerance2 Example \(VB.NET\)](#)

#### See Also

[IDimXpertSymmetryTolerance Interface](#)  
[IDimXpertSymmetryTolerance Members](#)

#### Availability

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**Modifier Property (IDimXpertSymmetryTolerance)**

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertSymmetryTolerance Interface](#) : Modifier Property (IDimXpertSymmetryTolerance)

Gets the material condition modifier for this Dimxpert symmetry tolerance.

#### .NET Syntax

Visual Basic (Declaration)

ReadOnly Property Modifier As [swDimXpertMaterialConditionModifier\\_e](#)

Visual Basic (Usage)

 Copy Code

```
Dim instance As IDimXpertSymmetryTolerance  
Dim value As swDimXpertMaterialConditionModifier\_e  
  
value = instance.Modifier
```

C#

```
swDimXpertMaterialConditionModifier\_e Modifier {get;}
```

C++/CLI

```
property swDimXpertMaterialConditionModifier\_e Modifier {  
    swDimXpertMaterialConditionModifier\_e get();  
}
```

#### Property Value

Material condition modifier as defined in [swDimXpertMaterialConditionModifier\\_e](#)

#### Visual Basic for Applications (VBA) Syntax

See [DimXpertSymmetryTolerance::Modifier](#).

#### Example

[Get DimXpert Tolerance2 Example \(VBA\)](#)  
[Get DimXpert Tolerance2 Example \(VB.NET\)](#)

#### See Also

[IDimXpertSymmetryTolerance Interface](#)  
[IDimXpertSymmetryTolerance Members](#)

#### Availability

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# IDimXpertTangencyTolerance (Type)

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IDimXpertTangencyTolerance Interface

See Also [Members](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertTangencyTolerance Interface

Allows you to access DimXpert tangency tolerances.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

 .NET Syntax

Visual Basic (Declaration)

Public Interface IDimXpertTangencyTolerance

Visual Basic (Usage) 

Dim instance As [IDimXpertTangencyTolerance](#)

C#

public interface IDimXpertTangencyTolerance

C++/CLI

public interface class IDimXpertTangencyTolerance

 Visual Basic for Applications (VBA) Syntax

See [DimXpertTangencyTolerance](#).

 Example

[Get DimXpert Tolerance6 Example \(VBA\)](#)  
[Get DimXpert Tolerance6 Example \(VB.NET\)](#)  
[Get DimXpert Features and Annotations in a Model Example \(C#\)](#)

 Remarks

Tangency tolerances are created automatically by the system when dimensions are created that constrain aspects of the part.

You can access more general geometric tolerance information using [IDimXpertTolerance](#).

 Accessors

[IDimXpertAnnotation::GetAppliedAnnotations](#) and [IDimXpertAnnotation::IGetAppliedAnnotations](#)  
[IDimXpertPart::GetAnnotations](#) and [IDimXpertPart::IGetAnnotations](#)  
[IDimXpertPart::GetAnnotation](#)  
[IDimXpertFeature::GetAppliedAnnotations](#) and [IDimXpertFeature::IGetAppliedAnnotations](#)  
[IFeature::GetSpecificFeature2](#)

 See Also

[IDimXpertTangencyTolerance Members](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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IDimXpertTangencyTolerance Interface Members

See Also [Properties](#)



[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertTangencyTolerance Interface

The following tables list the members exposed by [IDimXpertTangencyTolerance](#).

 Public Properties

| Name                                                                                                            | Description                                                               |
|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
|  <a href="#">OriginFeature</a> | Gets the DimXpert feature of origin for this DimXpert tangency tolerance. |

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 See Also

[IDimXpertTangencyTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**IDimXpertTangencyTolerance Interface Properties**

[See Also](#)

[SolidWorks.Interop.swdimxpert Namespace](#) : IDimXpertTangencyTolerance Interface

For a list of all members of this type, see [IDimXpertTangencyTolerance members](#).

 Public Properties

| Name                                                                                                            | Description                                                               |
|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|
|  <a href="#">OriginFeature</a> | Gets the DimXpert feature of origin for this DimXpert tangency tolerance. |

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 See Also

[IDimXpertTangencyTolerance Interface](#)  
[SolidWorks.Interop.swdimxpert Namespace](#)

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**OriginFeature Property (IDimXpertTangencyTolerance)**

[See Also](#) [Example](#)



[SolidWorks.Interop.swdimxpert Namespace](#) > [IDimXpertTangencyTolerance Interface](#) : OriginFeature Property (IDimXpertTangencyTolerance)

Gets the DimXpert feature of origin for this DimXpert tangency tolerance.

 .NET Syntax

Visual Basic (Declaration)

ReadOnly Property OriginFeature As DimXpertFeature

Visual Basic (Usage)  [Copy Code](#)

```
Dim instance As IDimXpertTangencyTolerance
Dim value As DimXpertFeature

value = instance.OriginFeature
```

C#

```
DimXpertFeature OriginFeature {get;}
```

C++/CLI

```
property DimXpertFeature^ OriginFeature {
    DimXpertFeature^ get();
}
```

**Property Value**

[IDimXpertFeature](#)

 Visual Basic for Applications (VBA) Syntax

See [DimXpertTangencyTolerance::OriginFeature](#).

 Example

[Get DimXpert Tolerance6 Example \(VBA\)](#)  
[Get DimXpert Tolerance6 Example \(VB.NET\)](#)  
[Get DimXpert Features and Annotations in a Model Example \(C#\)](#)

#### See Also

[IDimXpertTangencyTolerance Interface](#)  
[IDimXpertTangencyTolerance Members](#)

#### Availability

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